

Reliability-weighted target prioritization in CD4+ T-cell Perturb-seq: a generalizability-theory decomposition

Che Cheng¹

¹Institute of Statistical Science, Academia Sinica, Taipei, Taiwan.
che830621@stat.sinica.edu.tw. ORCID 0000-0003-3376-7833.

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Abstract

Genome-wide Perturb-seq screens prioritize candidate targets by the strength of a perturbation’s transcriptional effect. Effect strength does not answer a prior measurement question: is the readout dependable? A large effect estimated from a single guide, a single donor, or a pseudobulk of few cells need not survive replication, and for target prioritization each false lead costs a validation experiment. We treat each perturbation effect as a measurement in a crossed Target $\times$ Guide $\times$ Donor $\times$ Condition design and apply generalizability theory (Cronbach et al., 1972) to separate the dependable part of an effect from facet-specific idiosyncrasy. Guides and donors enter as random facets; condition enters as a fixed facet and is analyzed within its levels. For each target we report a dependability profile over the facets and a joint generalizability coefficient over the two random facets, and we re-rank targets by effect magnitude weighted by that coefficient. On the released screen (Zhu et al., 2025), removing the measurement-error floor estimated from the non-targeting controls raises the number of genes with a dependable target-signal share above .10 from 40 to 7,674. A design study indicates that reliability is limited by the number of guides rather than the number of donors. Analyzed within activation states, dependability recovers the T-cell-receptor signaling module as reliably measurable only in activated cells, without recourse to gene annotation. Every methodological decision was recorded and adversarially reviewed, and all results regenerate from the released summary statistics.

1 Dependability, not effect size

Genome-scale Perturb-seq couples pooled CRISPR perturbation with a single-cell transcriptomic readout, so that the transcriptional consequence of perturbing each of thousands of genes is measured in one experiment (Dixit et al., 2016; Replogle et al., 2022). Applied to primary human CD4+ T cells, the cells that coordinate adaptive immunity and are a substrate for engineered cell therapies, such a screen identifies the genes whose perturbation most reshapes a T-cell program and, on that basis, nominates candidate targets (Zhu et al., 2025). The nomination carries a cost. Each target that advances is committed to an arrayed knockdown, a functional assay, or an animal model, so a hit that does not reproduce spends a validation experiment on an artifact. What a screen is worth therefore depends not only on the effects it detects but on whether the ranking it induces over targets can be trusted.

Target prioritization in Perturb-seq conventionally ranks candidates by differential-expression strength: effect size, z -score, false-discovery rate, or the number of downstream differentially expressed genes. These summaries answer whether a perturbation moved the transcriptome *in the observed dataset*. They do not answer whether the effect would hold up under another guide, another donor, or another cell-state context. The two questions come apart. A target can carry a large apparent effect that is fragile, when two guides disagree, one donor drives the signal, knockdown is weak, or an off-target flag is present; a target of moderate effect that is consistent across guides and donors can be the better validation bet. We therefore treat *dependability* as a first-class quantity alongside magnitude.

Two properties are at stake that effect size does not by itself deliver: *accuracy*, that a reported effect is dependable rather than measurement noise, and *generalizability*, that it recurs across guides, donors, and cell-state contexts. We therefore ask three questions, at two levels of the measurement. *RQ1 (per target)*: which targets carry an individually dependable effect, one that would recur under a fresh guide and a fresh donor, rather than a large but fragile one? *RQ2 (the design)*: how dependable is the screen as a whole, at its two guides and four donors? *RQ3 (improving the design)*: which facet limits that dependability, so that a future screen knows whether to add guides or donors? RQ1 is the per-target coefficient on which we rank targets; RQ2 and RQ3 are design-level questions, answered by projecting the same variance decomposition to alternative designs. Answering any of them first requires separating the dependable part of an effect from measurement noise.

This concern is not new in high-throughput biology, though it has rarely been framed as measurement reliability. Li et al. (2011) call discoveries by their reproducibility across replicates rather than by effect size, defining an irreproducible discovery rate as an analogue of the false-discovery rate; Billmann et al. (2023) show that standard CRISPR-screen quality metrics do not by themselves capture the reproducibility of context-specific hits; and data-driven target-prioritization frameworks integrate several weighted evidence types rather than fitness effect alone (Behan et al., 2019). Generalizability theory supplies the missing piece: a single interpretable coefficient for the dependability of a per-target effect, together with an explicit account of which experimental facet limits it.

The rest of the paper develops the per-target coefficient and the two levels it rests on (Section 2), describes the data and the pre-registered choice of estimator (Section 3), and takes up RQ1 through RQ3 across four analyses (Section 4).

2 A generalizability-theory decomposition

2.1 The crossed design and its facets

The experimental structure maps onto a psychometric measurement design (Lord & Novick, 1968). The target gene is the object of measurement; the true perturbation signature is the universe score; the two guides per gene act as parallel forms; the four donors and the three culture conditions are crossed facets; cell sampling and sequencing contribute residual error. Classical test theory writes an observed score as $X = T + E$, its true score the expected observation over independent replications (Novick, 1966); generalizability theory (Cronbach et al., 1972) decomposes the error term into named sources,

$$E = E_{\text{guide}} + E_{\text{donor}} + E_{\text{condition}} + E_{\text{interactions}} + E_{\text{residual}},$$

and reports how much of the effect is attributable to each. We distinguish random from fixed facets. Guides and donors are *random* facets: one wants the effect to generalize across the guides

and donors that were not sampled. Condition is a *fixed* facet: Rest, Stim8hr, and Stim48hr are three deliberately chosen activation states, not a sample from a population of conditions, so the appropriate treatment is a separate analysis within each state rather than a generalization across states. A perturbation that acts only in the activated state is context-specific, not unreliable, and folding condition into a generalization coefficient would misclassify that signal as noise.

2.2 Per-target dependability

For target t we summarize dependability as a *profile* of three facet coefficients, reported separately and never multiplied. Each observation is a (target, guide, donor, condition) pseudobulk carrying an effect profile over the 18,129 genes, defined relative to a matched non-targeting-control reference. For a set of observations, the group-mean profile is the elementwise mean, and each facet coefficient is the Pearson correlation, across genes, between two such mean profiles that differ only in that facet:

- R_t^{guide} : the target’s guides are split into two halves, each half’s mean profile is formed, and the two are correlated. A high value indicates that the transcriptional signature reproduces across independent guide sets.¹
- R_t^{donor} : the same construction with a two-versus-two donor split, followed by a moment-based empirical-Bayes shrinkage toward the pooled mean, which is warranted because four donors leave the donor coefficient with three degrees of freedom and correspondingly noisy.
- $R_t^{\text{condition}}$: the mean of the pairwise correlations among the three within-state profiles, reported as cross-context *consistency* rather than as a generalization coefficient.

The ranking scalar is the per-target dependability $R_{\text{dep},t}$, the joint generalizability over the two random facets. Writing each facet coefficient as $R = \tau/(\tau + e)$ for a shared true-score variance τ and a facet-specific error variance e , the error-to-signal ratios $(1 - R)/R = e/\tau$ add, so

$$R_{\text{dep},t} = \frac{1}{1 + \frac{1 - R_t^{\text{guide}}}{R_t^{\text{guide}}} + \frac{1 - R_t^{\text{donor}}}{R_t^{\text{donor}}}}, \quad S_t = E_t R_{\text{dep},t}, \quad (1)$$

where E_t is the effect magnitude and S_t the dependability-weighted score used for ranking. We keep the naming discipline that separates the two levels of the theory: $R_{\text{dep},t}$ is a per-target quantity, whereas the design-level coefficients $E\rho^2$ and Φ describe the measurement design as a whole and carry no per-target subscript.

The joint coefficient has a direct interpretation. Writing an observed profile as $\tau_t + \varepsilon_{\text{guide}} + \varepsilon_{\text{donor}}$, the correlation between two *independent replications* of the whole measurement, a second experiment with fresh guides and fresh donors, is $\sigma_\tau^2/(\sigma_\tau^2 + \sigma_{\text{guide}}^2 + \sigma_{\text{donor}}^2)$, which is Equation (1). The single-facet coefficients are the same correlation with only one facet refreshed; the joint coefficient refreshes both and is therefore the most stringent form of reproducibility. The equality is exact under the additive model; a guide-by-donor interaction would add a further disagreement term, so the error-adding form is an upper bound when interactions are present.

¹These are raw single-observation split-half correlations. The reliability of the deployed two-guide average is the Spearman–Brown step-up $2r/(1 + r)$, a separate and larger quantity; we report the single-observation form because the joint coefficient is defined at the single-guide-by-single-donor level. Both are distribution-light empirical correlations rather than Gaussian variance-component ratios.

3 Data and estimation

The data are the released summary statistics of a genome-scale Perturb-seq screen in primary human CD4+ T cells (Zhu et al., 2025): CRISPR perturbations of 18,129 genes in a crossed design of two guides per gene, four donors, and three culture conditions (Rest, Stim8hr, Stim48hr), with non-targeting guides as controls. We work from the released joint pseudobulk and define each perturbation’s per-gene effect as its non-targeting-control-relative log-CPM within the matched donor-condition-run stratum, so that the effect is measured against the controls that share its technical context. No single-cell data are processed.

The gene-level variance components are estimated with a precision-weighted crossed mixed model fit per gene, following the generalizability-theory design of Section 2 (Brennan, 2001). The estimator was not assumed. Three candidates were compared by pre-registered synthetic recovery, and the precision-weighted model was the only one to recover the planted components in both Gaussian and heavy-tailed regimes; a distance-based candidate failed, because a distance share is not a variance component.

4 Results

4.1 Measurement error masks dependable signal

The residual of a pseudobulk decomposition combines biological idiosyncrasy with measurement error, and the latter is inflated when a pseudobulk aggregates few cells. The non-targeting controls carry no perturbation, so the variance of the non-targeting effect within its stratum estimates the per-gene measurement-error floor directly. Subtracting this floor from the residual raises the median target-signal share from .019 to .081 and raises the number of genes with a target-signal share above .10 from 40 to 7,674, or 42% of the genome (Figure 1). The dependable signal was present but masked; the same controls give a clean non-targeting-versus-non-targeting negative test ($r \approx 0$), so the correction introduces no signal where none exists.

4.2 A guide-limited design

A design study projects the generalizability coefficient onto alternative numbers of guides and donors (Figure 2). On the representative signal genes the current design (two guides, four donors) reaches $E\rho^2 \approx .44$. The guide error term exceeds the donor error term by a factor of about 2.4; adding one guide raises the coefficient about three times as much as adding one donor. With two guides the coefficient is capped near .53, and no number of donors reaches .70, whereas .70 is attainable with about 15 guides. Guide-to-guide variability, rather than donor heterogeneity, is the binding constraint, which is actionable guidance for the design of subsequent screens. Repeating the design study within each activation state returns the same verdict in all three states (Table 1), so the recommendation is not an artifact of pooling; within a state the guide term dominates the donor term more sharply still, because most donor disagreement is a cross-state phenomenon.

4.3 A reliability-weighted ranking

Ranking targets by S_t reorders the candidate list relative to effect size alone (Figure 3). Of the 100 highest-effect targets, 49 fall out of the 100 most dependable, and reproducible targets that effect size underranks rise (Table 2). These are the targets a dependability-weighted screen would nominate for validation but a strength-ranked screen would overlook.

Removing measurement error via the non-targeting controls

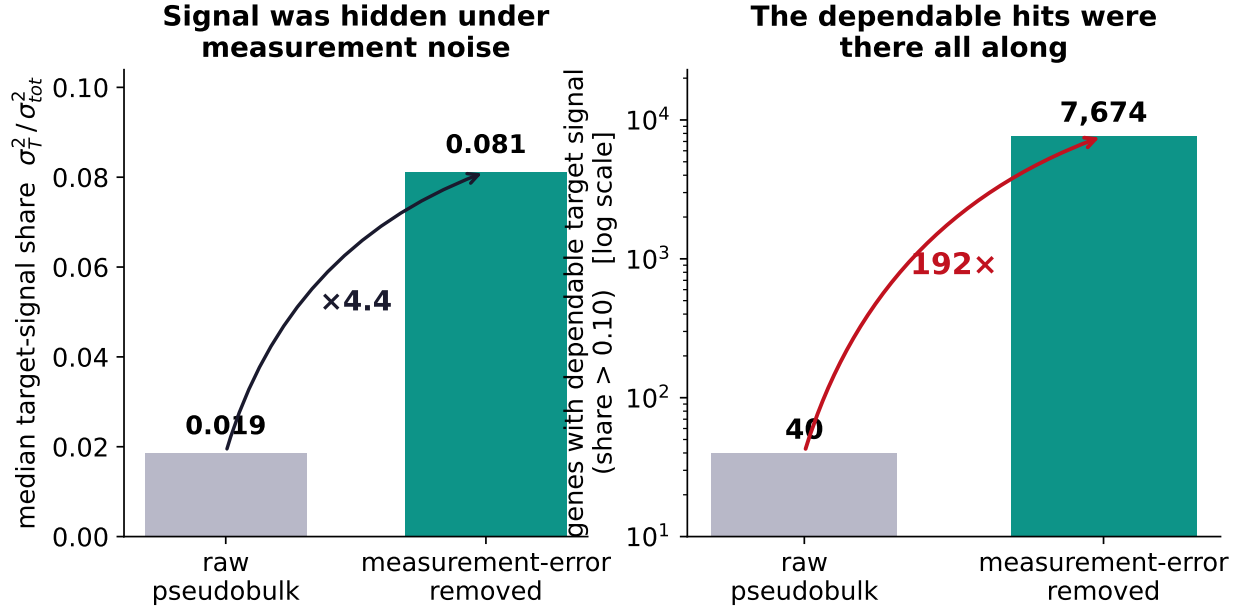


Figure 1: Measurement-error removal via the non-targeting controls. Left, the median target-signal share; right, the number of genes with a dependable target-signal share above .10, on a logarithmic axis.

4.4 Context-specific dependability recovers the TCR module

Analyzed within activation states, dependability separates targets that are reliable across all states from targets that are reliable in one state and not another (Figure 4). Because a within-state coefficient rests on roughly a third of the cells and runs lower than the pooled value, states are compared to one another rather than to the pooled number. One hundred targets are context-specific in this sense. Among them, the T-cell-receptor (TCR) signaling module, the CD3 complex (CD3D, CD3G, CD247) with the proximal kinase ZAP70 and the scaffold LAT, is reliably measurable only in activated cells, with resting dependability at the noise floor (resting $R_{dep} \approx .02-.05$; stimulated $\approx .20-.28$). The knockdown effect for CD3D grows with activation (effect magnitude 0.26, 0.57, and 1.12 across Rest, Stim8hr, and Stim48hr) and becomes dependable only upon activation. The framework reconstructs a coherent signaling module from dependability alone, without gene annotation, which is the strongest available evidence that it measures a real property of the perturbation rather than an artifact of the summary.

5 Reproducibility

The pipeline runs end to end from the released summary statistics, without processing the full single-cell data: data retrieval, a checksummed evidence manifest, identifiability gates, synthetic method selection, per-gene decomposition, measurement-error removal, the design study, and the per-context ranking. All results are committed as tables and figures, and every reported number

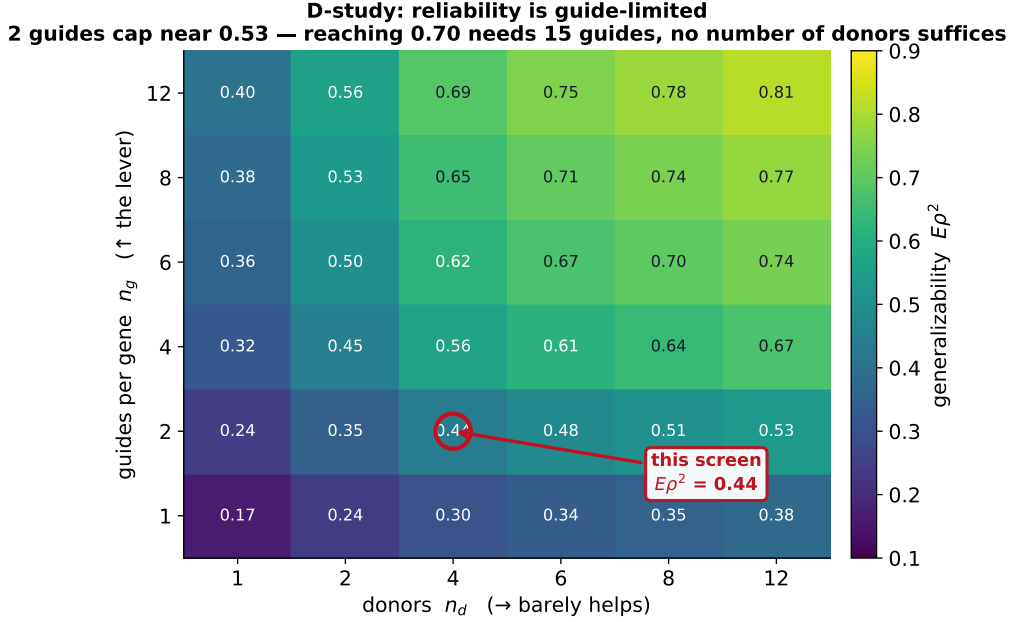


Figure 2: The generalizability coefficient $E\rho^2$ across the design grid. The current screen (two guides, four donors) is marked; reaching .70 requires about 15 guides and is unreachable by adding donors.

Table 1: Per-condition design study on the representative signal genes. The guide-limited verdict holds within every activation state.

Activation state	$E\rho^2$	Guide error	Donor error
Rest	.502	0.026	0.005
Stim8hr	.507	0.023	0.004
Stim48hr	.491	0.019	0.004
Pooled	.438	0.040	0.017

regenerates from a single re-run. The code is released under Apache-2.0.

6 Limitations

Four donors leave the donor variance component with three degrees of freedom, so the donor coefficient and the donor axis of the design study carry real uncertainty, which we report as bands rather than point values. Sequencing run is partially confounded with donor, a fact of the released design that we record rather than correct. A small number of full-scale Monte-Carlo gate confirmations are scheduled on a compute cluster; the results shown are the local synthetic and real-data tier. Finally, this is a re-analysis of released data, so the design-study guidance is prospective, addressed to subsequent screens rather than to the present one.

A A recorded, cross-model-hardened process

The methodology was developed under an auditable process (Figure 5). Each decision (the choice of estimator, the thresholds, the classification of each facet as random or fixed, and the criterion for

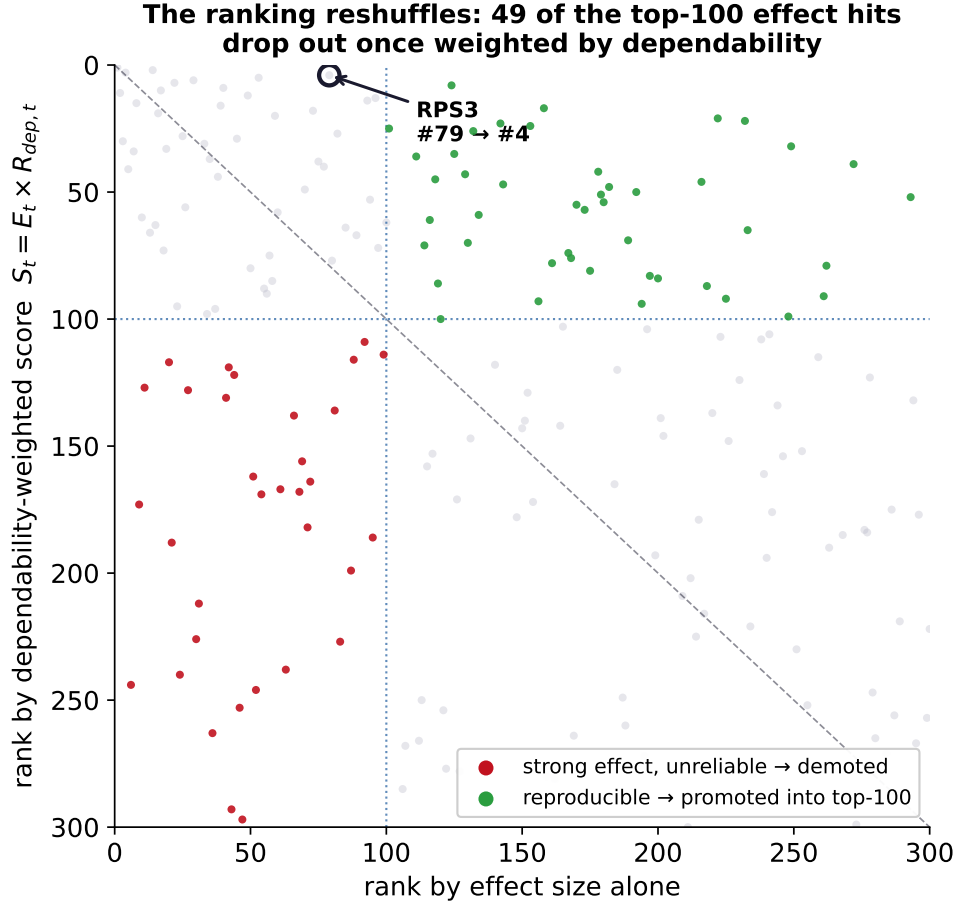


Figure 3: Rank by effect size alone (horizontal) against rank by the dependability-weighted score (vertical). Points below the diagonal in the lower-left quadrant are demoted (strong but unreliable); points in the upper right are promoted (reproducible).

identifiability) was recorded as a versioned issue using issue-driven development (Cheng, 2026a) and spec-driven development, both open-source tooling the author maintains. Because the reasoning was documented rather than implicit, the frozen record could be subjected to adversarial review by an independent frontier model, which returned a blocked verdict with 11 findings, three of them critical, spanning identifiability, measurement error, selection, validation leakage, and compute. Each finding was resolved within the same recorded loop: the falsification gates and controls were frozen before any real result was inspected, the estimator was selected as in Section 3, and a mis-specified reliability aggregation (a product of coefficients) was replaced by the joint coefficient of Section 2.2. The resolutions were then cross-verified across models with a second adversarial pass and with `parallel-ai-agents` (Cheng, 2026b), a package the author maintains that dispatches a task to independent agents and cross-compares their outputs. The recorded decisions, the blocked verdict, and every resolution are versioned in the repository.

Table 2: Targets that effect size underranks but dependability promotes. Rank is out of the 7,169 scored targets.

Target	Rank by effect	Rank by dependability	$R_{\text{dep},t}$
RPS3	79	4	.34
QARS1	53	5	.32
RPP21	124	8	.34
IMP4	96	13	.31
NMD3	158	17	.33

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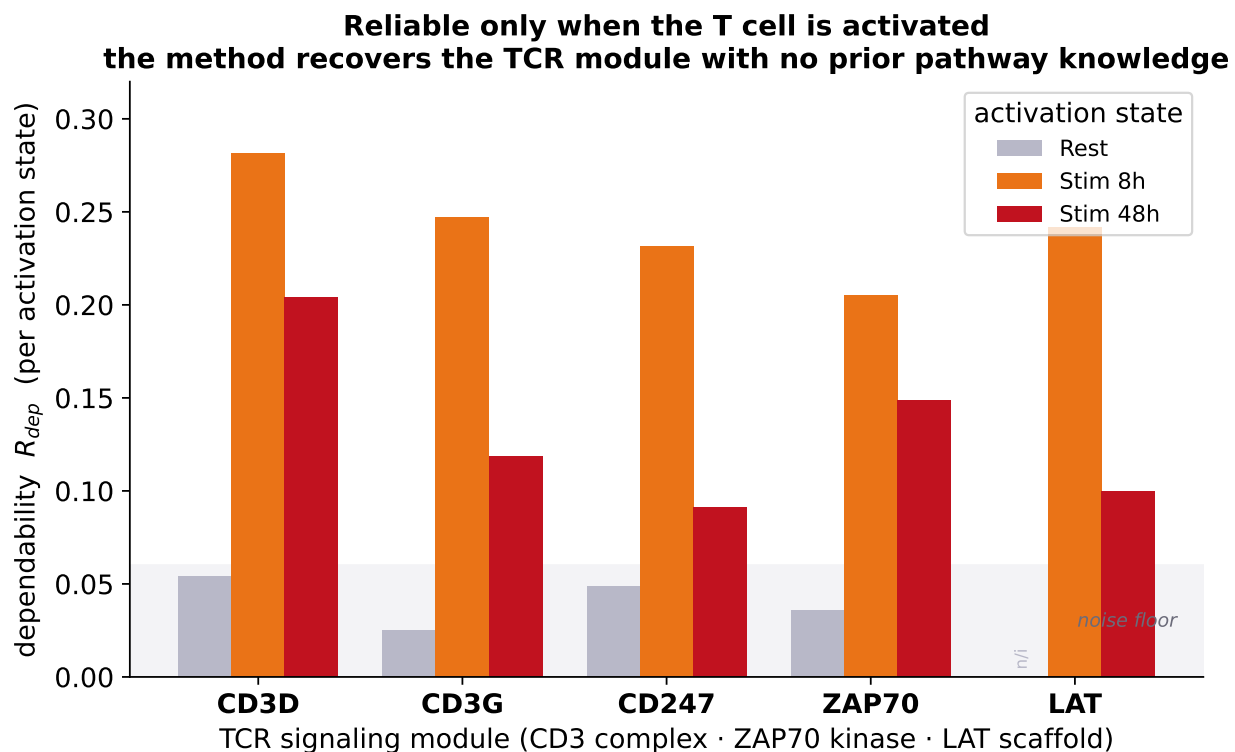
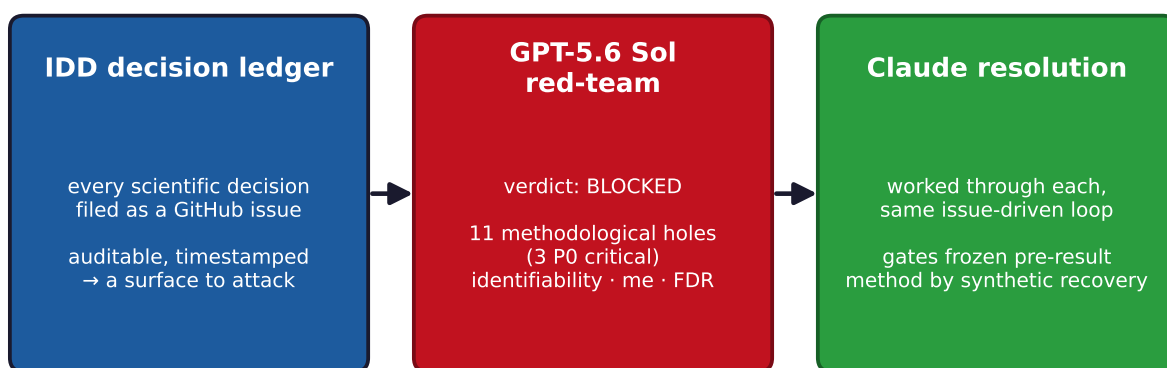


Figure 4: Per-state dependability for the T-cell-receptor signaling module. Resting values sit at the noise floor; the effect becomes reliably measurable only upon activation.

An auditable, cross-model-hardened process



decide & document → get torn apart → repair (never “certified” — hardened)

Figure 5: The recorded, cross-model-hardened process: a decision ledger enables adversarial review by an independent model, whose blocked verdict is resolved in the same loop. The verdict was blocked and worked through, not certified.